Comprehensive Risk Register & Threat Matrix — WariMitra

Document Identifier: WM-RSK-020

Classification: Project Risk Assessment & Threat Mitigation Strategy

Standard: ISO 31000 / BABOK v3 Section 6.3

Target Audience: Program Managers, Solution Architects, Lead Business Analysts, Interview Panels

Evidence Sources: `Reference files/backend_audit.md:54-63`, `ARCHITECTURE.md:474-484`, `docs/SECURITY.md`

1. Risk Analysis Methodology

Risks are quantified using a 5×5 Risk Severity Matrix:

$$\text{Risk Severity} = \text{Probability (1–5)} \times \text{Impact (1–5)}$$

- **Critical (16–25):** Severe threat to life safety, complete system outage, or statutory breach.
- **High (10–15):** Impaired operational capability; significant delay in emergency response.
- **Medium (5–9):** Moderate operational friction; acceptable workaround available.
- **Low (1–4):** Cosmetic or minor administrative inconvenience.

2. Risk Register

Category A: Risks Identified Directly from Project Codebase & Audits

Risk ID	Risk Classification	Risk Description & Cause	Operational Impact	Prob (1–5)	Imp (1–5)	Severity Score	Mitigation Implemented in Codebase	Risk Owner	Current Status
RSK-001	`IDENTIFIED`	**Database**	Total system	5	5	**25 (Critic**	Pivoted backen	Solution	**Mitigat**

	`FIED`	**se**	system			**(Critic**	backen	n	**Mitigat**

Risk ID	Risk Classification	Risk Description & Cause	Operational Impact	Prob (1–5)	Imp (1–5)	Severity Score	Mitigation Implemented in Codebase	Risk Owner	Current Status
	`FROM PROJECT`	**Lockups from Concurrent GPS Pings:** Monolithic relational database locks up when 100k+ pilgrims send periodic location updates simultaneously	freeze; emergency SOS alerts blocked.			**al)**	d to Firebase Cloud Firestore and serverless Cloud Functions, auto-scaling to 10k+ writes/sec (`ARCHITECTURE.md:553`).	Architect	**ed**

neously

Risk ID	Risk Classification	Risk Description & Cause	Operational Impact	Prob (1–5)	Imp (1–5)	Severity Score	Mitigation Implemented in Codebase	Risk Owner	Current Status
		(`backend_audit.md:58`).							
RSK-002	`IDENTIFIED FROM PROJECT`	**Fake SOS Flooding / Denial-of-Service:** Malicious actors or bots flood emergency API with fake panic requests (`backen`	Responders overwhelmed; genuine life-threatening emergencies delayed.	4	5	**20 (Critical)**	Implemented 3-second hold physics gesture, device idempotency UUIDs, 5 req/min rate limiting, and India-only geofence bounds	Lead Backend Dev	🩶 **Mitigated**

		(backen					bounds		

Risk ID	Risk Classification	Risk Description & Cause	Operational Impact	Prob (1–5)	Imp (1–5)	Severity Score	Mitigation Implemented in Codebase	Risk Owner	Current Status
		d_audit.md:59, functions/src/index.ts:59).					.		
RSK-003	IDENTIFIED FROM PROJECT	**Blockchain RPC Node Congestion / Outage:** Public Ethereum testnet RPC times out or is rate-limited	Failure to record cryptographic proof of emergency events.	4	3	**12 (High)**	Implemented automated fail-safe fallback: Cloud Function catches RPC failure, stores hash in local Firestore with	Web3 Lead	🩶 **Mitigated**

limited

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Risk ID	Risk Classification	Risk Description & Cause	Operational Impact	Prob (1–5)	Imp (1–5)	Severity Score	Mitigation Implemented in Codebase	Risk Owner	Current Status
		during high-frequency emergency events (`docs/WEB3_ARCHITECTURE.md:32`).					status `LOCAL_PROOF_ONLY`.		
RSK-004	`IDENTIFIED FROM PROJECT`	**Open Test Security Rules in Firestore:** Firestore security rules for `sosAle`	Data tampering or unauthorized read of sensitive incident coordinates.	4	4	**16 (Critical)**	Must reinstate strict role check `canManageEmergencies()` prior to official production pilgrim	Security Analyst	**Open / In Progress**

		`sosAle`					pilgrim		

Risk ID	Risk Classification	Risk Description & Cause	Operational Impact	Prob (1–5)	Imp (1–5)	Severity Score	Mitigation Implemented in Codebase	Risk Owner	Current Status
		`rts` and `blockchainProofs` temporarily open with `allow read, write: if true;` for rapid testing (`firestore.rules:55,180`).					age deployment.		

Category B: BA-Recommended Operational & Environmental Risks

Risk ID	Risk Classification	Risk Description & Cause	Operational Impact	Prob (1–5)	Imp (1–5)	Severity Score	Recommended Mitigation Strategy	Risk Owner	Current Status
RSK-005	`BA-RECOMMENDED RISK`	**Cellular Blackouts in Rural Ghat Corridors:** Pilgrimage passes through remote valleys (e.g., Dive Ghat) where cellular towers experience 100% capacit	Mobile apps cannot reach Cloud Functions or Firestore.	4	4	**16 (Critical)**	Pre-cache regional maps and static clinic directories on device; implement background SQLite retry queue syncing upon signal re-acquisition.	Mobile Lead	◍ **Planned**

Risk ID	Risk Classification	Risk Description & Cause	Operational Impact	Prob (1–5)	Imp (1–5)	Severity Score	Recommended Mitigation Strategy	Risk Owner	Current Status
		capacity saturation.							
RSK-006	`BA-RECOMMENDED RISK`	**Device Battery Depletion on Long Marches:** Pilgrims walk 10–12 hours daily without charging points; continuous GPS polling exhaust	Devotee phones die, leaving them untraceable and unable to call for help.	4	3	**12 (High)**	Implement adaptive location throttling: reduce location ping frequency to every 5 minutes when device battery drops below 20%.	Product Owner	◍ **Recommended**

exhaust

Risk ID	Risk Classification	Risk Description & Cause	Operational Impact	Prob (1–5)	Imp (1–5)	Severity Score	Recommended Mitigation Strategy	Risk Owner	Current Status
		s phone batteries.							
RSK-007	`BA-RECOMMENDED RISK`	**Demographic Digital Literacy Hesitation:** Elderly rural pilgrims (70+ years) mistrust smartphone apps and rely strictly on	Low mobile adoption in the most vulnerable demographic.	4	3	**12 (High)**	Empower youth Ground Volunteers with the app to act as digital proxies and intermediaries on behalf of elderly devotees.	Community Lead	🟨 **Operationalized**

Risk ID	Risk Classification	Risk Description & Cause	Operational Impact	Prob (1–5)	Imp (1–5)	Severity Score	Recommended Mitigation Strategy	Risk Owner	Current Status
		on							
		physical inquiries.							